Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/13/2015
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2013	**End Date:**	11/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AB74G		
Performance Goal No.:			
Performance Goal Text:			
	Continuation of Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events (ELFSTONE) ; grant NNX10AV27G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany		

Task Description:	Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Conducted Facility Checkout and performance testing (FCE 1.1) for ISS levitation processing of Ti and FeCrNi in vacuum with successful sample positioning limits imposed 23-28 February 2015. Continued Performance testing on Vit106A, FeCrNi, FeCo, NiTa, CuCo 20-22 April 2015 with successful FeCrNi recalescence detection, double recalescence monitoring, oscillation triggering, and sample stirring control. Conducted planning for flight support ESL FeCrNi quenching and CoSi property measurement tests at ESL-MSFC on 17-28 August 2015. Conducted final three CuCo undercooling experiments to complete Batch 1.1 using MSL-EML on the ISS, 18 August 2015. MSL-EML IWG#11, German Space Agency (DLR), Köln Germany, 8-9 January 2015, planning of additional facility checkout testing (FCE 1.1) to confirm operational safety during heating and pulse excitation of samples on-orbit. MSL-EML IWG#12, German Space Agency (DLR), Köln Germany, 11-12 May 2015. Reviewed FCE results and success of operational changes; evaluated dust budget for Batch 2 and proposed samples for consideration for Batch 3. MSL-EML IWG#13, German Space Agency (DLR), Köln Germany, 8-9 December 2015. conducted Batch 1.1 science review and recommended on-board operational procedure changes for space testing. Served as Symposium Organizer for Properties for Materials Science at High Temperature 19th NIST (National Institute of Standards & Technology) Symposium held in Boulder CO on 21-26 June 2015 and American Society for Gravitational and Space Research (ASGSR) annual meeting to be held in Alexandria VA on 11-14 November 2015. Plenary Speaker for International Symposium on Physical Sciences in Space (ISPS)-6 Symposium, held on 15 September in Kyoto, Japan. (Ed. Note 12/10/2015: Continued by "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events--NNX16AB59G" ; grant NNX16AB59G; period of performance 12/1/2015-11/30/2018)
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Lee J, Rodriguez JE, Hyers RW, Matson DM. "Measurement of density of Fe-Co alloys using electrostatic levitation." Metallurgical and Materials Transactions B. 2015 Dec;46(6):2470-5. Published First online: 14 August 2015. http://dx.doi.org/10.1007/s11663-015-0434-7 , Dec-2015
Articles in Peer-reviewed Journals	Rodriguez JE, Matson DM "Thermodynamic modeling of the solidification path of levitated Fe–Co alloys." Calphad. 2015 Jun;49:87-100. http://dx.doi.org/10.1016/j.calphad.2015.03.001 , Jun-2015
Articles in Peer-reviewed Journals	Lee J, Xiao X, Matson DM, Hyers RW. "Numerical prediction of the accessible convection range for an electromagnetically levitated Fe50Co50 droplet in space." Metallurgical and Materials Transactions B. 2015 Feb;46(1):199-207. First online: 17 September 2014. http://dx.doi.org/10.1007/s11663-014-0178-9 , Feb-2015

Dissertations and Theses	Minervini F. "Development of a System for Radiation-based Thermophysical Property Measurement." MS Thesis, Tufts University, Medford MA, August 2015. , Aug-2015